

Developing and Evaluating a Website to Guide Older Adults in Their Health Information Searches: A Mixed-Methods Approach

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Arlene Fink¹ and John C. Beck²

Abstract

This mixed-methods study developed and evaluated an online program to improve older adults' skills in identifying high-quality web-based health information. We conducted focus groups and individual interviews to collect data on older adults' preferences for online instruction and information. We used the findings to develop, pilot test, and evaluate an interactive website which was grounded in health behavior change models, adult education, and website construction. Sixty four participants were randomly assigned to Your Health Online: Guiding eSearches or to an analogous slide-based-tutorial and compared in their knowledge, self-efficacy, and program assessment. Experimental participants assigned significantly higher ratings of usability and learning to the new site than controls did to their tutorial although no differences were found in self-efficacy or knowledge. Experimental participants reported that participation was likely to improve future searches. Information is now needed to examine if such programs actually improve health searches, ehealth literacy, and health outcomes.

Keywords

older adults, mixed methods, web-based health information, online education

¹UxI, Los Angeles, CA, USA

²The Langley Research Institute, Pacific Palisades, CA, USA

Corresponding Author:

Arlene Fink, UxI, 1562 Casae Road, Pacific Palisades, Los Angeles, CA 90272, USA.

Email: afink@mednet.ucla.edu

Introduction

The Internet has been recognized for many years as an important mechanism for transforming medical care (Baker, Wagner, Singer, & Bundorf, 2003; Kassirer, 1995; Purcell, Wilson, & Delamothe, 2002; Silberg, Lundberg, & Musacchio, 1997). In 2000, 46% of American adults reported access to the Internet, and 25% of American adults looked online for health information. By 2012, 81% of U.S. adults used the Internet, and 59% reported looking online for health information in the past year (Fox & Duggan, 2013).

According to the American Association of Retired Persons (AARP), 40% of adults 50 years of age or older feel very or extremely comfortable using the Internet (Koppen, 2010), and nearly 80% of persons 55 to 64 years, and 60% of those 65 and older go online at least once a day (Keenan, 2009). Internet use among people 65 years of age and older has risen significantly from 2009 to 2012 (Zickuhr & Madden, 2012). The baby boomers are among the most prolific Internet users, and many of them are now 65 years or older. The Pew Internet and American Life Project, which refers to boomers as the "silver tsunami," classifies their online behavior as similar to that of people in their 30s (Fox, 2001). Like them, baby boomers go online to get information, buy products and do their banking (Pew Internet and American Life, 2002). Online health information use among the elderly is likely to grow as the baby boomers age and the proportion of older adults as a percentage of the U.S. population increases.

Aging is often accompanied by increasing health problems and increased physician visits (Schiller, Lucas, Ward, & Peregoy, 2012). Among the reasons that older adults see their physicians relatively frequently is that they often have multiple, long-term medical problems that require regular care. Over 83% of medicare patients have at least one chronic condition such as hypertension or diabetes, and 23% have five or more (Anderson, 2005). A survey of over 8,000 people, 35% of whom were 55 years of age and older found that respondents who reported having two or more chronic diseases were more likely to search for online health information than respondents who reported having no chronic disease (Bansil, Keenan, Zlot, & Gilliland, 2006).

Internet consumers tend to rely on information without being particularly concerned with important quality criteria such as the source of the information or the date it was published. A 2006 survey (Fox, 2006) of Internet users reported that they start at a general search engine when researching health and medical advice online. Just 14% of patients living with disability or chronic disease, many of whom are older adults, said they "always" check the source and date of the health information they find online, while another 18% said they do so "most of the time." Sixty-seven percent of patients with

chronic conditions said they check the source and date “only sometimes,” “hardly ever,” or “never.” (Marshall & Williams, 2006) found that the two most important indicators of quality that consumers used were the site sponsor’s authority and the use of plain language.

People use the Internet for health information because of the convenience of being able to search for information at any time, unlimited access to inexpensive information, the ability to tailor queries, and user anonymity when searching for subject-sensitive health information (Anderson, Rainey, & Eysenbach, 2003; Bansil et al., 2006; Cline & Haynes, 2001; Rice, 2006). An additional factor influencing Internet use has been the emergence of the patient self-care or partnership model in health care delivery (Barry & Edgman-Levitan, 2012; Oshima Lee & Emanuel, 2013). A key assumption of the model’s proponents is that all patients benefit from having access to quality health care information because it helps them to ask physicians better questions and facilitates joint decision-making. Evidence suggests that patients who ask questions, express opinions, and state preferences during physician office visits have measurably better health outcomes than those who do not (Corrigan, 2000; Dwight-Johnson, Unutzer, Sherbourne, Tang, & Wells, 2001; Mahler & Kulik, 1990). Moreover, recent surveys show that nearly all patients prefer to be offered choices and to be asked their opinions (Frosch, May, Rendle, Tietbohl, & Elwyn, 2012; Levinson, Kao, Kubly, & Thisted, 2005).

The Internet is unregulated, however, and consumers gain access to information without guidance from health professionals (Marshall & Williams, 2006). Research has raised concerns about difficult-to-interpret (Czaja, Sharit, & Nair, 2008), false, misleading, or incomplete online health information (Berland et al., 2001; Eysenbach, Powell, Kuss, & Sa, 2002; Meric et al., 2002). An overabundance of irrelevant or invalid information may place new burdens on health care professionals and detract from their ability to provide care efficiently. Strategies and programs to teach effective searching skills are needed (Taha, Sharit, & Czaja, 2009), but few are available especially for the elderly.

This mixed-methods study aimed to improve older adults’ knowledge of and confidence in identifying high-quality online health information by developing and evaluating a theory-based educational website using the experience of a multidisciplinary team of educators and health professionals and relying on community participation in program planning and evaluation. A growing literature suggests that involving members of the community in research which is designed to improve their health results in encouraging novel ideas and approaches, and in facilitating intervention development and community buy-in (Horowitz, Robinson, & Seifer, 2009). Researchers in one

study, for instance, developed a mobile-enabled web application to promote physical activity for older cancer survivors and concluded that involving stakeholders in program planning and design was critical in enhancing program relevance and appeal and also in matching the needs of target users (Yan Hong et al., 2013). Similarly, researchers found that including older people in a senior residential community center was essential in developing an evidence-based program to enhance health literacy (Aspinall, Beschnett, & Ellwood, 2012). The website in this study is designed for adults who are 50 years of age and older who currently use the Internet, and it incorporates their preferences for online learning and health content.

Method

Study Design

This two-phase study was done in conjunction with a nonprofit, health and social services organization with deep roots in the Westside and the surrounding communities of Los Angeles. All participants were 50 years of age or older and had gone online for health information at least once in the past year. They were recruited for both study phases through posters, the organization's newsletters and website, local newspaper articles, and personal recommendations. Participants were reimbursed financially.

The study's first phase aimed to develop and pilot test the website. As part of development, we identified older adult's needs and preferences for online learning through focus groups and individual in-person interviews. Eligible focus group and interview participants were willing to be recorded and meet with study investigators for approximately 90 min. The focus groups took place at the community organization's main office and were led by the study's geriatrician. They were conducted in a semistructured format to allow participants to bring up topics that mattered to them and to build conversation using comments from the other group members. Each session had two note takers and was recorded. Each individual interview was conducted by the study's geriatrician and took place at the participant's home or the study's primary office. Interviewees did not attend the focus groups. The interviews were also recorded.

The principal investigator, website developer and graphic artist created a prototype of Your Health Online using focus group and interview results to guide the content selection and format. Other study team members and a consultant independently evaluated the prototype's usability, appearance, and potential usefulness. We then pilot tested the feasibility of using a prototype of Your Health Online. A feasible website can be used without supervision.

Pilot test participants were willing to spend up to 90 min to complete an online educational program and answer questions in person or via email about their experience.

The study's second phase consisted of a comparative evaluation of Your Health Online and an online tutorial prepared by the National Library of Medicine (NLM). Participants were eligible if they were willing to spend up to 70 min to complete an online educational program and questionnaire at a location of their choosing about their experience. We identified potential study participants from the community organization's list of users and volunteers. People who agreed to participate were screened by the organization's CEO, and if eligible, were assigned at random to an experimental or control group by the Pew Internet (PI) using a table of random numbers. Experimental group participants received a user name and password and were asked to complete a questionnaire that was embedded in the education program at its conclusion. Control group participants were directed to the NLM's website. They were also asked to complete a questionnaire, the link for which was included in the email asking for study participation. Questionnaire responses were automatically entered into the study's data base. We removed all identifying information from the database within 1 week of the conclusion of data collection.

The study protocol was approved by The Langley Research Institute's Institutional Review Board (IRB: 00005539).

Interventions

Experimental Program: Your Health Online: Guiding eSearches: The conceptual framework for the education was the Health Belief Model (HBM) which provides a framework for identifying personal (perceptions of severity and benefit, barriers to change, self-efficacy) and situational factors (cues to action) likely to influence health behavior (Glanz, Rimer, & Lewis, 2002; National Institutes of Health, 2006). Table 1 describes how each of HBM's change strategies was operationalized for use in Your Health Online. For example, the HBM recommends that change requires explaining to clients how to take action and what the positive results will be. Your Health Online applied this recommendation in at least three ways: providing user-friendly guidelines for evaluating website quality, providing a list of high-quality websites, and illustrating the value of using high-quality websites.

Your Health Online is also grounded in principles of learning and curriculum development, particularly Knowles' theory, which he called "andragogy." We supplemented Knowles' theory with the ADDIE framework of curriculum development because it implicitly incorporates Knowles' theory

Table 1. *Health Belief Model's Application to Your Health Online.*

Health belief model concept	Definition	Potential change strategies	Example application to your health online
Perceived susceptibility	Beliefs about chances the problem is applicable to the individual	Define what populations are at risk and their levels of risk Tailor risk information to an individual's characteristics or behaviors	People 50 years of age or older who use the Internet to review online health information Demonstrate potential hazards of relying on search engines; not distinguishing between content and ads
Perceived severity	Beliefs about the seriousness of the problem and its consequences	Specify the consequences of the problem and recommended action	Illustrate how time-consuming and wasteful a misguided search can be
Perceived benefits	Beliefs about the effectiveness of taking action to reduce risk or seriousness	Explain how, where, and when to take action and what the potential positive results will be	Provide user-friendly guidelines for evaluating website quality. Provide a list of high-quality web sites Demonstrate the efficiency and value of high-quality websites
Perceived barriers	Beliefs about the material costs of taking action	Correct misinformation	Emphasize through illustration, the benefits of the guidelines for evaluating quality
Cues to action	Factors that activate "readiness to change"	Provide "how to" information	Provide checklists and links Provide an interactive website with practice exercises
Self-efficacy	Confidence in one's ability to take action	Provide training and guidance in performing the action	Give numerous examples Incorporate content and tips from others on how to get the best information

of andragogy and advocates evaluation (Dick & Carey, 1996; Leshin, Pollock, & Reigeluth, 1992).

We used Visual Studio 2010 and SQL server 2008 to build an interactive website with password authentication for access. The site was hosted on a secure service provider under Internet Information Services (IIS7).

Comparison Program: Evaluating Internet Health Information: A Tutorial from the NLM: We chose this tutorial because it addresses many of the same topics that we anticipated covering in the new program, and it was developed

by a prestigious organization likely to be accepted as valid by the study's participants. The tutorial differed from our planned program in that it is not interactive (no practice exercises, feedback, links to other sites, ability to choose topics of interest at will) and it was not designed specifically for older adults. Instruction depends upon a series of slides and a narrator to explain each slide. In addition, to view the tutorial, the user must install the Flash plug-in, version 8 or above.

Data Collection

Focus Groups. The focus groups were designed to elicit information about community-dwelling older adult's knowledge of and experience online. To understand knowledge and experience, components of the HBM (Becker, 1974; Harrison, Mullen, & Gree, 1992; Janz, Champion, & Strecher, 2002), we asked questions like: Which websites do people currently use for health searches? Which topics are most important? How do they evaluate the accuracy of the information they obtain? We showed the groups *Evaluating Internet Health Information* and asked for a critique of the most and least favorable characteristics. We also addressed potential barriers to using the Internet, another HBM variable, and to online learning by asking questions like: How important are large font sizes, graphics, and links in selecting a site? We were also concerned with methods for improving self-efficacy (Bandura, 1997), particularly with respect to patient-physician communication, and asked questions about ways that valid information can assist patients in working together with physicians and other health professionals in managing their own health care.

Interviews. The individual interviews had the same purposes as the focus groups, but they permitted us to delve more deeply into individual perceptions. We developed a standard set of questions, but participants answered them in their own words. The open-ended questions asked whether participants used a specific website to begin their search or a search engine; what topics they researched; how frequently they went online for health information; and how they evaluated the quality of health websites and health information. We asked for advice about the contents and format of a website that would be useful and appealing. Interviewees were also asked to view *Evaluating Internet Health Information* and assess its usefulness, usability, appropriateness, and appeal.

Website Prototype Review. The study team agreed that a formal review was essential before pilot testing the prototype of *Your Health Online*. The review

was guided by the Accessible Health Information Technology (IT) for Limited-Literacy Populations Checklist, a Checklist for Developers and Purchasers of Health IT developed by the Agency for Healthcare Quality and Research (AHRQ; Eichner & Dullabh, 2007). Reviewers, who were senior team members, answered yes or no to items such as the following: presence of a great deal of white space (fewer words or less dense text), short line length (40-50 characters), and bullets to break up text; use of dark text on light background; ability to run site without requiring Flash, Shockwave, or other plug-ins; and ability to display and operate site on all major browsers. The review process was collaborative and iterative. We resolved disagreements through negotiation.

Pilot Test. We developed a “paper-and-pencil” questionnaire that asked participants if they learned anything new and to rate the site’s appearance and usability. We also asked participants to tell us if the exercises were useful. All questions were accompanied by a 10-point scale. For instance, we asked participants to rate the sites appearance from 1 = *terrible* to 10 = *excellent*, and to rate usability on a scale of 1 = *definitely not useful* to 10 = *definitely useful*. These data were used to improve the website but were not included in the evaluation.

Evaluation. All participants completed a 10-minute online survey with questions on demographics; knowledge of standards for identifying and evaluating online health websites; confidence in using the Internet for health information, experience with the Internet; and assessment of the usability, appearance, and usefulness of each of the two programs.

Knowledge. To measure knowledge, all participants were asked 8 true-false questions specifically developed for the study. Sample questions included: “The information on a website is probably created by the organization that built the site (*false*).” “If you visit a website that you trust, you can also trust that the links on that site will lead to trustworthy sites (*false*).”

Self-efficacy. All participants were asked 6 questions to measure their self-efficacy. Four of the six questions were adapted from the Pew Internet and American Life project. (Pew Internet and American Life, 2013) A sample question was “How confident are you that you can find valid online health information?” The response choices ranged from extremely to not very confident.

Internet use. We asked participants 12 questions on Internet use and information-seeking. These questions were also adapted from the Pew Internet

and American Life project (Pew Internet and American Life, 2013). Sample questions were, (a) "How often do you usually go online for health information?" The response choices ranged from once a week or more to less than once a month. (b) "The last time you went online for health information, how did you start?" The response choices were "with a search engine like Google or Yahoo" and "with a website I know."

Program assessment. We assessed perceptions of the program by asking all participants to tell us if their participation in the study will change the way they now do online searches (make it better or worse); how likely they are to use the information in their next search (definitely likely to definitely not likely); how much they learned (10 = *a great deal* to 1 = *nothing or almost nothing*); and whether the site is usable (10 = *extremely easy* to 1 = *extremely difficult*). We also asked for ratings of the experimental and control site's appearance (10 = *excellent* to 1 = *terrible*). We asked experimental participants to rate the usefulness (definitely useful to definitely not useful) of the exercises and interactive links to health-related websites. All participants were invited to comment on the study and the interventions.

Data Analysis

Two members of the study team reviewed the qualitative data obtained from the Phase 1 focus groups, in-person interviews, and pilot test. Each had access to recordings and notes. Study team members were asked to review the data for common themes. They then met in a day-long session to discuss and analyze their findings for each study component. The evaluation study phase relied on quantitative methods. We computed frequencies and percentages for the survey responses. We used the χ^2 or Fisher's Exact to test for differences between groups, and we report the exact *p*-values. We defined statistical significance as $p \leq .05$.

Results

Focus Groups. Nearly all attendees in the two focus groups were retired, although 5 had part-time paying jobs. The first group had 8 participants (5 females) and the second had 10 participants (6 females) for a total of 18. Several major themes emerged: The majority of participants used a search engine as an initial portal for health information, and none had heard of Medline Plus. Many participants first checked several sites, and if agreement was found among them, assumed that the information was reliable. Few compared the sites' accuracy. Focus group participants consistently asked for a

list of high-quality sites. When it came to pedagogy, nearly all participants agreed that a useful website should take between 30 to 60 min to complete and contain practice quizzes. Links to websites should be included that refer people to information on chronic disease management and medical procedures. The participants unanimously agreed that an interactive website was preferable to the NLM's slide presentation, which was perceived as old-fashioned and condescending.

Interviews. We conducted individual in-person semistructured interviews with six people (4 females). All participants were asked the same questions, but they were free to respond in their own words. Five of the interviewees were baby boomers, and all were employed. This group was much more skeptical of the web than either of the focus groups. For example, one interviewee stated that it was practically impossible at the present time to avoid giving up some of your privacy in exchange for a "free" web. According to this interviewee, most people do not read privacy policies. He also indicated that he was aware, that one of his most frequently used sites often contained outdated information.

Five of the six interviewees went to the web primarily when they had specific reasons (e.g., to learn about their own or a loved one's serious illness). In the course of their research, they became familiar with several high-quality sites including the National Cancer Institute and the Mayo Clinic. Two people relied on their health plan's site for information. None of the interviewees had heard of the NLM or Medline Plus. Five of the six described the NLM's tutorial as inadequate.

All six interviewees agreed that when they went to the web for health information, they tended to do so uncritically. For instance, they did not check when the site was last updated or if the content was original or came from another site. They were unanimous in their support of an educational website that was interactive, providing links to high-quality information, and offering practice in applying knowledge to problems like those they might encounter in their next searches.

Pilot Test. Four people (2 women) completed the pilot test. All participants reported learning new information and planned to use the information in the future. They found the exercises useful and the site usable. The appearance of the site ranged from 6 to 8, on a scale with 10 = *excellent* to 1 = *terrible*. Pilot test participants valued the exercises, with at least one person suggesting that the site include more. Participants indicated that several of the site's features were particularly helpful including the provision of list of high-quality websites and information on how to critically read a website and distinguish fact

from advertising. We used the focus group “themes,” interviews and pilot test findings to help guide program development. The program, for example, was designed to include links to high-quality websites, especially those that contain information on chronic disease management, medication, and medical procedures because these are of concern to older people.

Evaluation. Experimental and control groups were similar in sex and age, with an average age of 68.5 years. Participants ranged in age from 56 to 101 years. Just over 60% of the sample was female. 77% had a college education or greater. Participants in both the experimental and control groups achieved an average score of about 80% on a 10-item knowledge test. The control group was significantly more likely to incorrectly answer “true” to the statement, “Using a search engine is the best way to find reliable online health information” than the control (83.3% vs. 66.7% respectively), and to incorrectly answer “true” to the statement, “The information on a health website is probably created by the organization that built the site” (63.3% to 15%).

Table 2 shows that participants in both groups (77.8% and 70%) stated they were extremely or very confident in finding valid online health information and agreed that their searches helped them to ask their doctors new questions (72.2% and 76.7%)

Experimental and control group participants were similar in the frequency and purpose of the search as well as whether they searched for themselves or others (Table 3). More than half of all study participants use the web for information about doctors or other health professionals but not for hospital or other medical facilities. On average, 74% of study participants began their searches with a general search engine, with 24.2% of the study participants beginning searches using a specific website. Nearly all participants in both study groups visited between two and four sites each time they search for health information, with 47.7% of experimental and 43.3% control participants indicating that following the advice they found was a moderate or major help. About 80% of the control and 90% of the experimental group stated that participation in the study will make their searching better, but the 10-point difference was not statistically significant. Over 90% of both groups reported that they would probably or definitely use the information from the study in their next search.

Participants in both groups agreed that involvement in the study will make their searching better, and that they definitely or probably will use the information when they do their next search (Table 4). When asked how much they learned on a scale of 1 to 10, with 10 representing the best score, there was a statistically significant difference between groups, with the control assigning more low ratings (1's and 2's). Similarly, the experimental group assigned

Table 2. Self-Efficacy.

	Guiding eSearches (experimental) N = 36		Evaluating online health information (control) N = 29		P
	n	%	n	%	
Extremely or very confident in finding valid information (N = 35)	28	77.8	21	70.0	0.4718
Agree or strongly agree: Helped me to speak to doctors and other health professionals	20	57.2	19	63.3	0.6115
Agree or strongly agree: Helped me ask my doctor new questions	24	72.2	23	76.7	0.7198
Agree or strongly agree: Helped me in treating an illness or condition	18	50	17	56.7	0.5890
Agree or strongly agree: Helped me maintain my health or someone else's	22	61.1	20	66.7	0.6404
Agree or strongly agree: Affected my decision to see a doctor	16	44.4	8	26.6	0.1349

significantly more high ratings (9's and 10's) to their site's usability than did the control. Over 90% of the experimental group reported that they found the exercises to be definitely or probably useful.

We received three comments from three experimental program participants. Two stated that they benefited from the study, while the third recommended that the site developers consider adding an audio option. Two control participants also commented. Both indicated that the content was useful; one stated that the narrator spoke too slowly, and that the program needed to be updated and made interactive.

Conclusion

We used a mixed-methods approach to design and evaluate the feasibility of Your Health Online: Guiding eSearches, an online educational program to teach older adults to identify and assess high-quality health information websites. Thirty people participated in the development of the website, and 64 provided information on the new program's content and, format. When compared to a noninteractive tutorial with similar objectives developed by the NLM, participants assigned significantly higher ratings to the experimental

Table 3. Internet Use and Experience.

	Guiding eSearches (experimental) N = 36		Evaluating online health information (control) N = 30		P
	n	%	n	%	
Frequency of online searches					
Once a week or more	5	13.9	8	26.7	0.1937
Online is for self	19	52.8	19	63.3	0.3876
Reasons for search					
Specific disease or medical problem	34	94.4	30	100.0	0.4965
Certain medical treatment or procedure	30	83.3	27	90.0	0.4941
Exercise or fitness	22	61.1	16	53.3	0.5244
Doctors or other health professionals	20	55.6	18	60.0	0.7160
Prescription or over-the-counter drugs	23	63.9	18	60.0	0.7457
Hospitals or other medical facilities	14	38.9	10	33.3	0.6404
Health insurance	19	52.8	13	43.3	0.4446
Alternative treatments or medicine	17	47.2	13	43.3	0.7521
Depression, anxiety, stress, or other mental health concerns	12	33.3	13	43.3	0.4043
How search is started					
Search engine	25	71.4	24	80.0	0.4239
Specific website	10	28.6	6	20.0	
Number of sites visited					
2–3	15	41.7	18	60.0	0.4079
3–4	15	41.7	9	30.0	
How much help by following Internet advice					
Moderate or major help	17	47.2	13	43.3	0.7521
Don't know	2	5.6	5	16.7	0.2311
Participation in study will make searching better	33	91.7	24	80.0	0.2804
Definitely or probably use information from study in next search	34	94.4	28	93.4	0.9999

site's usability and how much they thought they had learned although no differences were found in self-efficacy or knowledge.

Among the possible reasons for the program's strengths was the active participation of prospective users in all study phases. In this study, we involved the community in focus groups, interviews, and the pilot test of Your Health Online. We used the information we gained from participants to

Table 4. Program Assessment.

	Guiding eSearches (experimental) N = 36		Evaluating online health information (control) N = 30		P
	n	%	n	%	
Participation in program will make searching better	33	91.7	24	80.0	0.2804
Definitely or probably use information from education in next search	34	94.4	28	93.4	0.9999
How much did you learn?					
A great deal	9	25.0	9	27.0	0.8300
Almost nothing	0	0.0	5	15.0	0.0211
Site's appearance					
Excellent	11	30.5	9	27.2	0.7640
Terrible	0	0.0	2	6.0	0.2251
Usability					
Extremely easy	27	75	12	40	0.0093
Extremely difficult	0	0.0			
Benefit from an optional voice-over					
Probably and definitely yes	16	72.2	NA		
Usefulness of exercises					
Definitely or probably useful	35	97.2	NA		

Note. $p \leq 0.05$

develop and revise the website's content and format so that it reflected their needs and preferences.

When developing Your Health Online, we adhered to the guidelines suggested for providing accessible information technology for people with limited literacy (Eichner & Dullabh, 2007). The National Assessment of Adult Literacy (National Center for Education Statistics, 2003) found that 71% of adults older than age 60 have difficulty using print materials, 80% have difficulty using documents such as forms or charts, and 68% have difficulty with interpreting numbers and doing calculations. Although this study's population was relatively well-educated, little is currently known about the relationship between aging, education, and ehealth literacy. Because of the lack of evidence, this study did not assume that education is associated with ehealth literacy. Therefore, in keeping with the guidelines, Your Health Online includes a great deal of white space, short line length, bullets, and questions and answers to break up the text (Eichner & Dullabh, 2007).

There are several limitations to this study. The findings come from a small, educated-volunteer sample of older adults who are current users of the Internet and have experience searching for online health information. We did

not collect information on health status or ethnicity, and these factors may have affected participants' perception of and willingness to use the website. As the Pew Internet and the American Life surveys have shown (Fox & Duggan, 2013), older adults use the Internet less frequently than do other age groups, and those who do, tend to have higher incomes and education than their peers. However, the smart phone may be altering the balance by improving access to and use of online health information for many, including the elderly and ethnic minorities, who are currently nonusers. A Pew survey found that among all cellphone users, Hispanics and African Americans are more likely than others to look for online health information on their phones (Fox & Duggan, 2012).

Low health literacy is a potentially serious problem among all older adults, many of whom have chronic illness and need to understand how to manage their health and navigate the health care system. We did not address this important topic because the study's sample size and duration were insufficient to examine the effects of health literacy on the quality of online searches, or the effects of high-quality Internet searches on health literacy. We think that understanding the relationship between health literacy and wise Internet use, or ehealth literacy is extremely important and should be the focus of future research.

This study had methodological limitations. We adapted measures of self-efficacy and Internet experience from the Pew Internet and the American Life project, assuming that they would be appropriate. We did not test this assumption; however, we did not receive complaints about the measures, nor did we discern any anomalies in our analysis of the responses. We also developed a 10-item knowledge test to measure the educational material's content, but we did not examine the extent to which good scores on the test predicted good searching skills. We cannot say with confidence that doing well on the test is indicative of skilled online health information searches.

Despite the study's limitations, it has several strengths. The study incorporated potential users' perspectives in the development of the website, in accordance with recommended guidelines for creating accessible health information technology (IT) for populations, such as older adults, with potentially low health literacy. Further, the education is grounded in accepted models of behavior change and adult learning. Finally, the evaluation's findings come from a systematic comparison with a freely accessible and suitable alternative program developed by the NLM.

Based on this study's results, we conclude that a website such as Your Health Online: Guiding eSearches is feasible for use among many older adults, and that it contains useful and useable information. People who work with older adults should consider how to obtain and integrate web-based instruction into their practices. Among an online program's beneficial

features are that it can be personalized (e.g., users may take as much time as they need), made interactive (e.g., users can receive immediate feedback), and updated regularly. To our knowledge, this is the first online educational program that combines these characteristics to meet the needs of the growing population of older Internet users. Future research should focus on evaluating the website's effectiveness in improving the way divergent groups of older adults, especially those of differing ethnicities and socioeconomic status, actually search for online health information and the effects of better searches on the quality and value of health care, offline and ehealth literacy, health outcomes, and quality of life.

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Author Biographies

Arlene Fink, PhD, is a senior health scientist at the Langley Research Institute and a Professor of Medicine and Public Health at UCLA. Dr. Fink is an expert in program evaluation and in the use of IT to promote better health care decision making. She has been responsible for developing an online alcohol screening and education system that is being used throughout the world to monitor alcohol-related risks in the elderly. In recent years, she has focused much of her research on promoting ehealth literacy in older adults.

John C. Beck, MD, is an emeritus Professor of Medicine at UCLA. He was the founder of UCLA's Multi Campus Division of Geriatrics and Gerontology, which, for two decades, has consistently been ranked among the top three Divisions of Geriatrics in the US. In recent years, he has devoted his research efforts to helping older adults navigate the health care system and maintain their independence.